

digital format, and may be gathering dust in some forgotten shelves. When these parallel datasets are digital, they may not have a standard encoding, such as Unicode, which makes it hard for machines to understand. Finally, these documents may not contain the informal language and slang used in social media posts or spoken dialects. Companies are finding creative ways to work around this problem.

Now it would just take too much time to train a model on every possible language pair. Google translate, for example, works on about 130 language pairs, of the 7000 languages in the world. To speed up the process, Google Engineers came up with what they call Zero Shot Translation. First, the neural network was trained to translate between Japanese and English, and then to translate between Korean and English. Then the translation knowledge itself was transferred to a pair that the neural network was not explicitly trained on, between Korean and Japanese. The neural net was embedding the context and meanings of the sentences in a representation that only it could understand, and which it used to translate between novel language pairs. Google Researchers refer to this internal language that emerged as “itnerlingua”. Microsoft Researchers used something they call a Universal Lexical Representation, which is a virtual map of meanings of words and sentences. By projecting the training data from multiple languages onto a single map, the machine can guess the meanings of words that it has not encountered before. The system can do this by finding similar representations from a shared space, and assigning a probabilistic value to the options. For example, “toamna”, the Romanian word for Autumn is weighted closer to “autumn” than say, “spring”. The system is smart enough to adapt to different situations depending on the needs of the translation. It can either look for similar words from languages in the same group, or languages that geographically overlap, without knowing the distinction between the two. This is because the system can

use the degree of overlap between the monolingual data sets. When translating from Romanian for example, it uses more data from Greek and Czech, less from German, and almost nothing from Finnish. When translating Italian, the system used other Romance languages such as Romanian as well as Czech, which overlaps with the region in which Italian is spoken, although it is not a Romance language.

MIT researchers came up with a similar approach as well, which require training from only monolingual language sets, without any examples of translations between two languages. The machines can be trained using unsupervised data sets, which means that there is no need for annotations and guidance. The words from each language are represented by an array of numbers, with a distance between them to show their relations. When maps of individual languages are made, patterns emerge that are common in all languages. For example, relations between families, the names of the months in a year, and the four directions appear as closely related clusters in all the languages. The distances between each of these words can also be matched to derive the meaning in another language. The actual translation involves applying statistical techniques to this array, or map of meanings, and choosing the option that is the most likely to be correct.

Researchers from Facebook used a similar approach as MIT to develop a machine translation system of their own. The system was trained in single languages using dictionaries, and the words were mapped to a two-dimensional space. Now the relative distances between the words are preserved, and the two-dimensional space only has to be rotated to correctly map the entire language space to another language. Facebook researchers used the simple method of checking the words that appeared around a particular word to build these spaces. This exploits the physicality of the world we live in, which is then described in a language. For example, the word “furry” is more likely to appear next

to the word “kitten”, as against the word “rocket”. To translate between the languages, all the machine needs to do is find the closest word in the corresponding target language. The approach also allows for rapid translation without the need for training based on language pairs.

Word-by-word translation is unreformable, and just not good enough for translating one language to another. The approach used by the Facebook developers is to use language modeling, to find out what makes fluent sentences within a single language. The models that identify fluent sentences in a single language, is trained on monolingual data, and does not need a corresponding translation from another language. The approach works for language pairs that are very different from each other (such as English and Russian), language pairs that have very few common resources (such as English and Romanian), as well as language pairs that are both low in resources and very different from each other (such as English and Urdu).

VOICE ASSISTANTS

When people originally imagined machine assistants for humans, before the advent of computers, no one imagined keyboards or mice as input devices. The most natural way to communicate was through voice. For the users who are getting their hands dirty with technology for the first time, and accessing the internet through touchscreens without even having used computers, voice is the most natural way for interactions. For example, India's new internet users prefer to listen to web pages rather than scrolling through them. Google Go has rolled out a feature meant specifically for these users, that allows users to listen to web pages in 28 Indian languages. The feature has been designed to work on spotty 3G connections as well.

The advancement of artificial intelligence has finally allowed for natural language processing and speech synthesis, but machines still have a hard time grasping the nuances of the spoken word. You cannot just ask Pho-